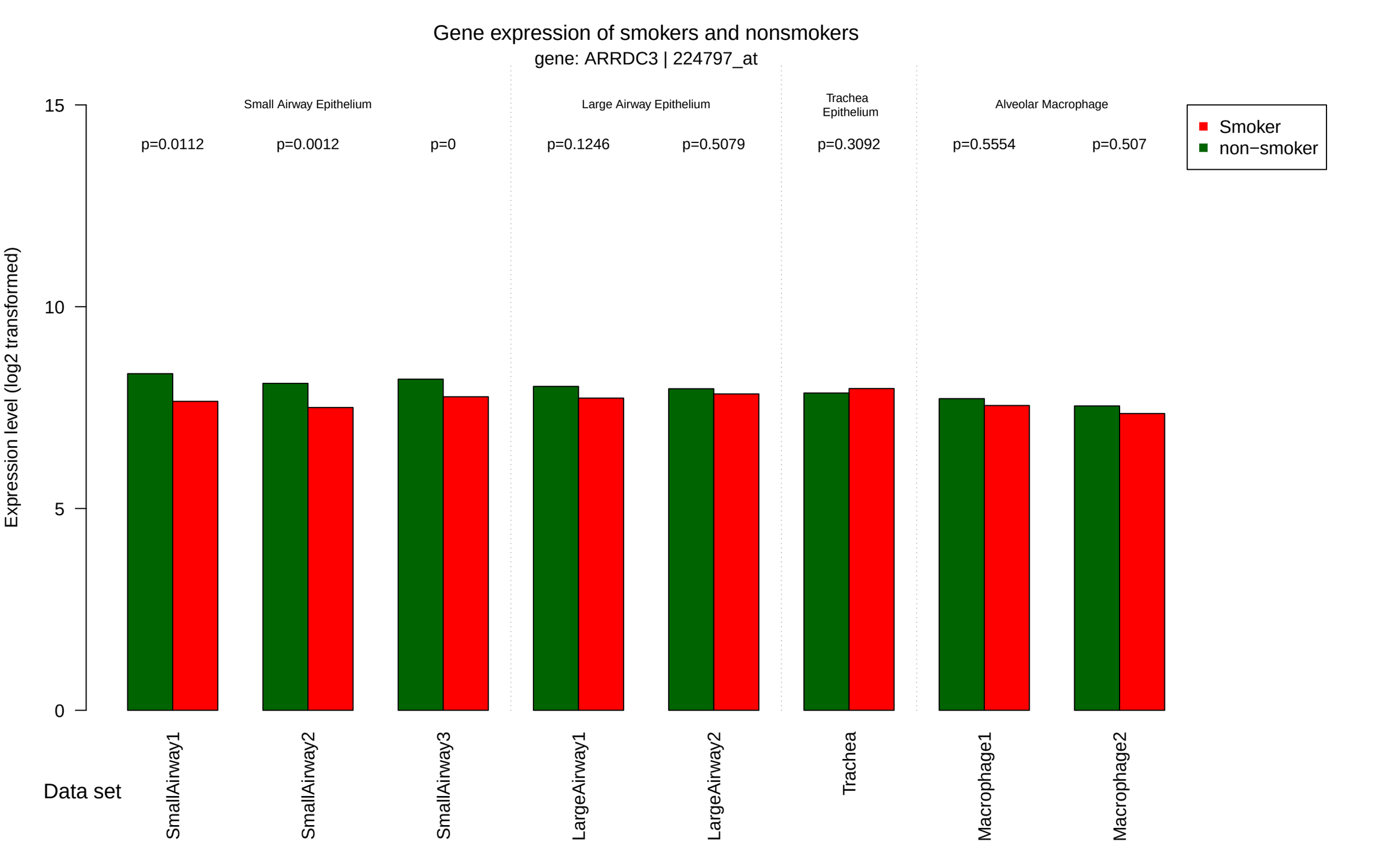
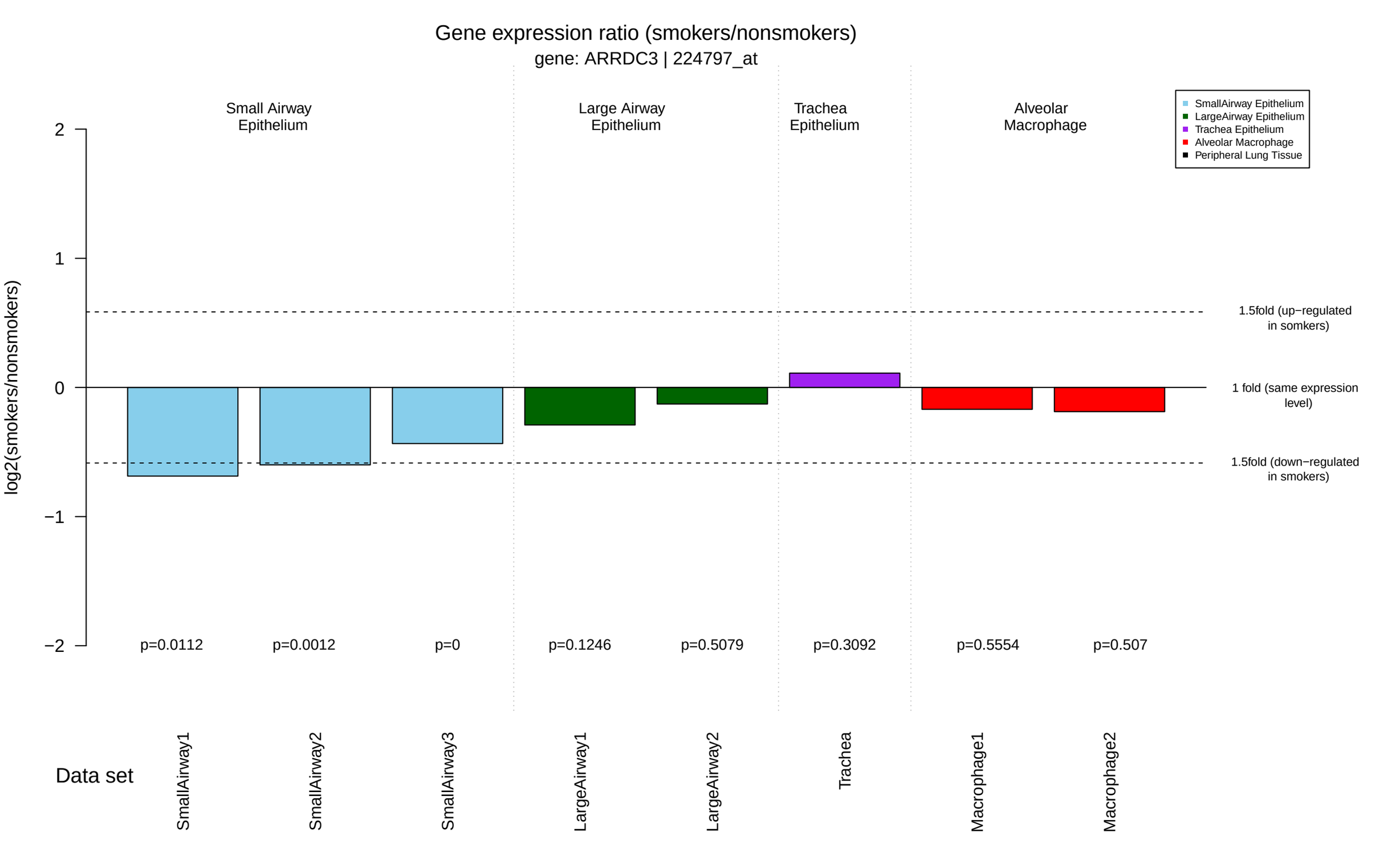
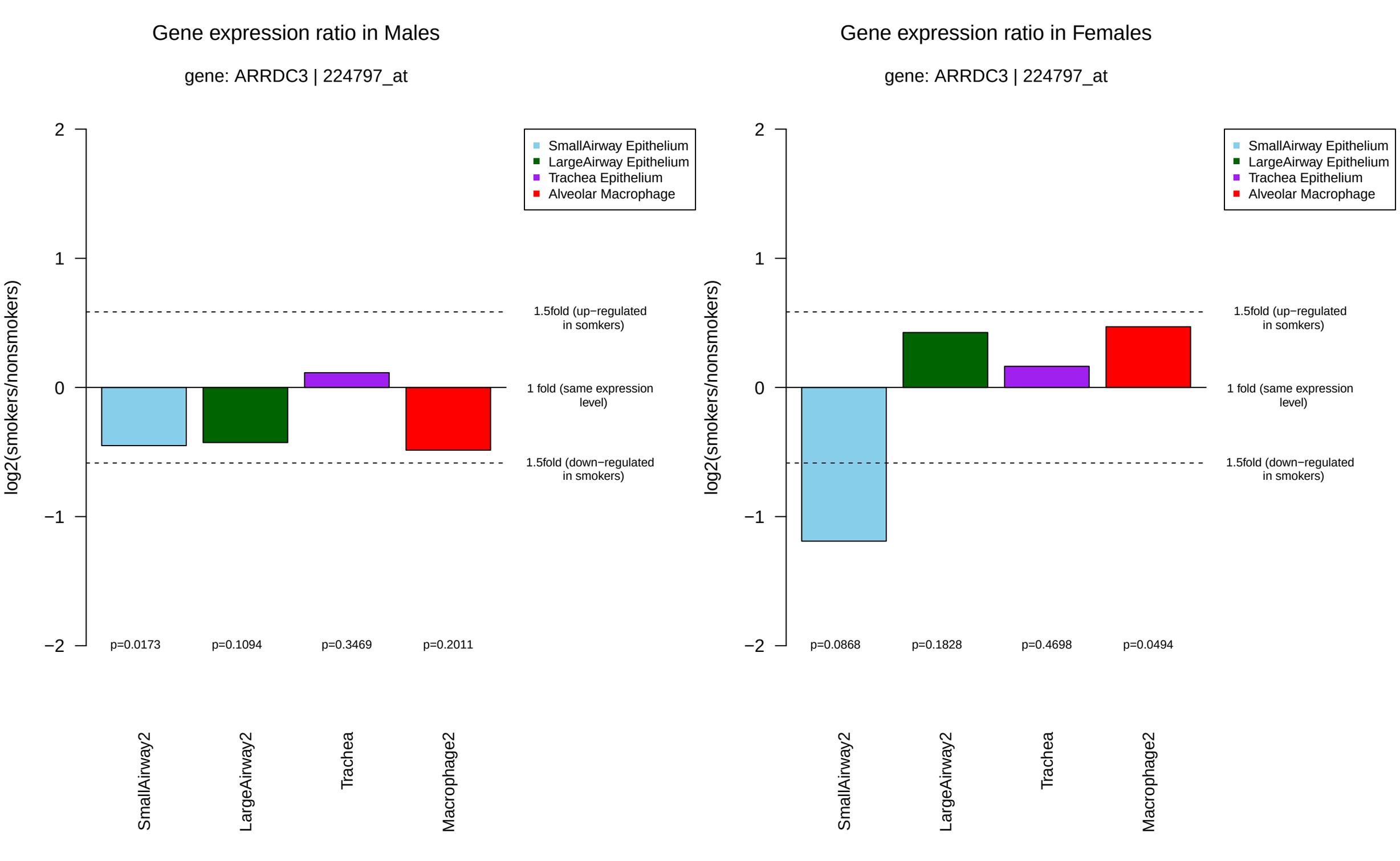
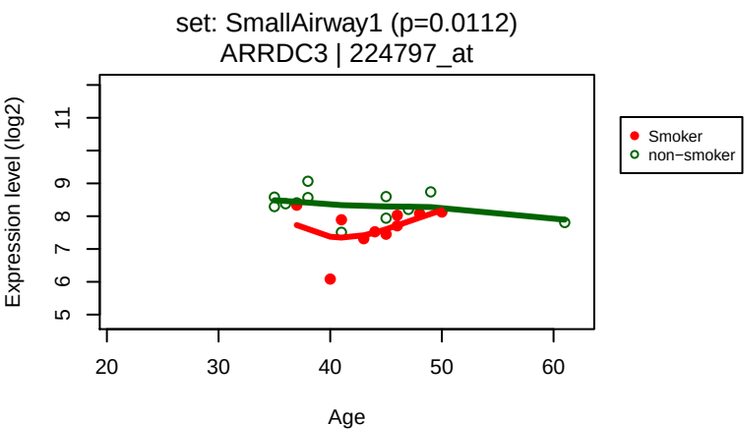
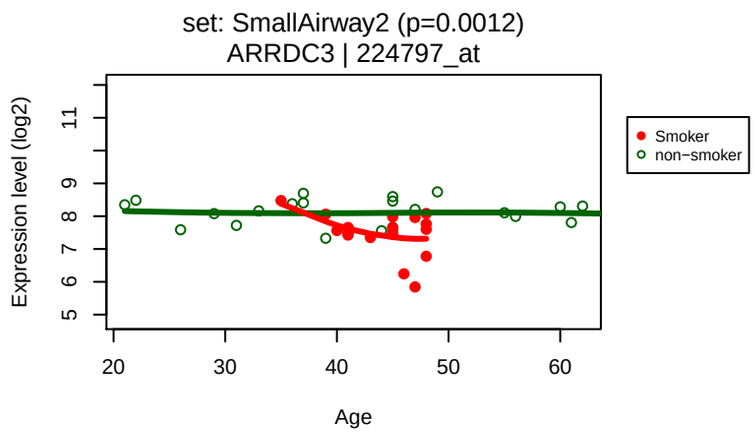
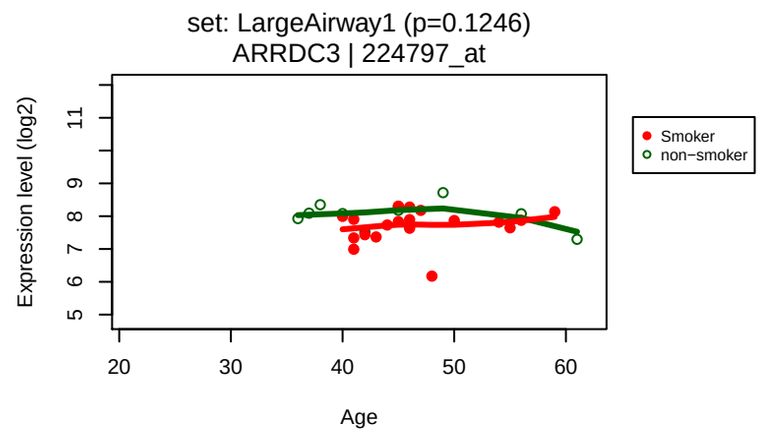

Gene expression of smokers and nonsmokers
gene: ARRDC3 | 224797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0112
p=0.0012
p=0
p=0.1246
p=0.5079
p=0.3092
p=0.5554
p=0.507
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

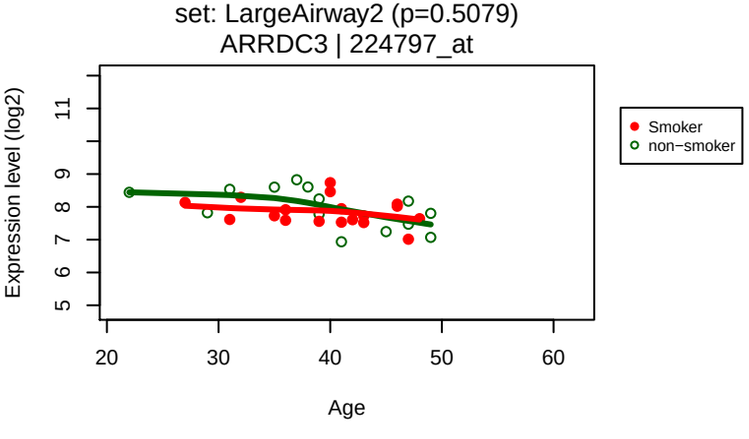

Gene expression ratio (smokers/nonsmokers)
gene: ARRDC3 | 224797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.0012
p=0
p=0.1246
p=0.5079
p=0.3092
p=0.5554
p=0.507
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

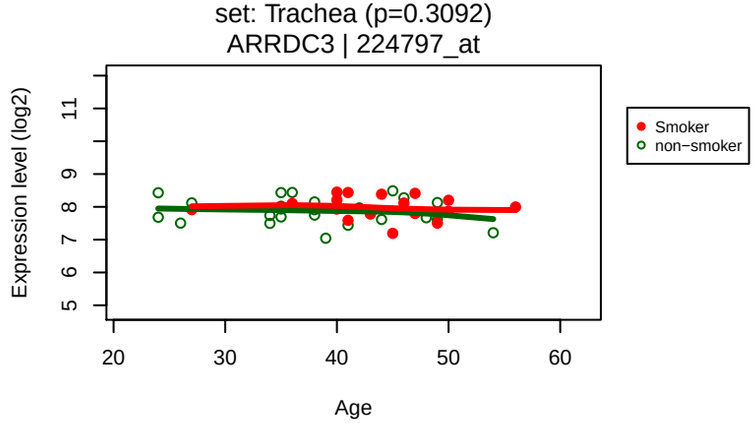

Gene expression ratio in Males
gene: ARRDC3 | 224797_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0173
p=0.1094
p=0.3469
p=0.2011
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARRDC3 | 224797_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0868
p=0.1828
p=0.4698
p=0.0494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

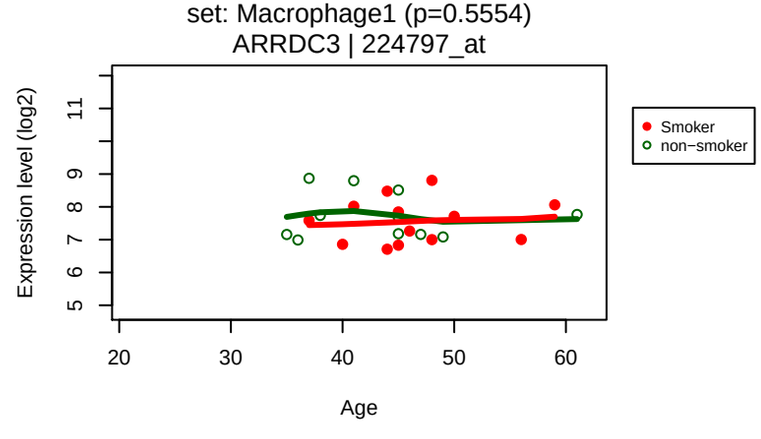

set: SmallAirway1 (p=0.0112)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker

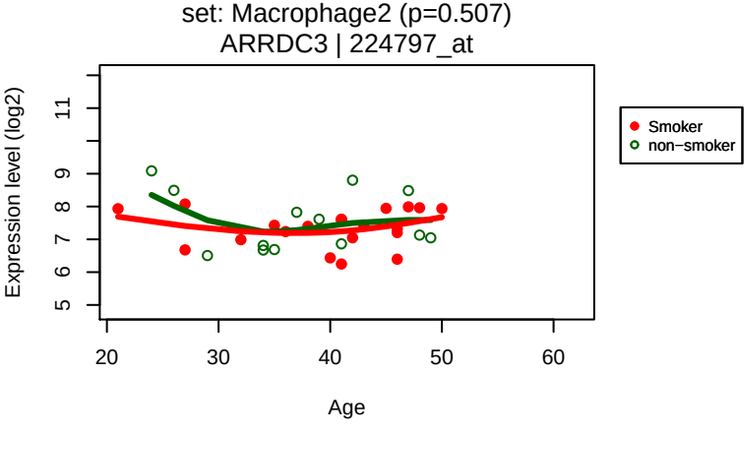

set: SmallAirway2 (p=0.0012)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker

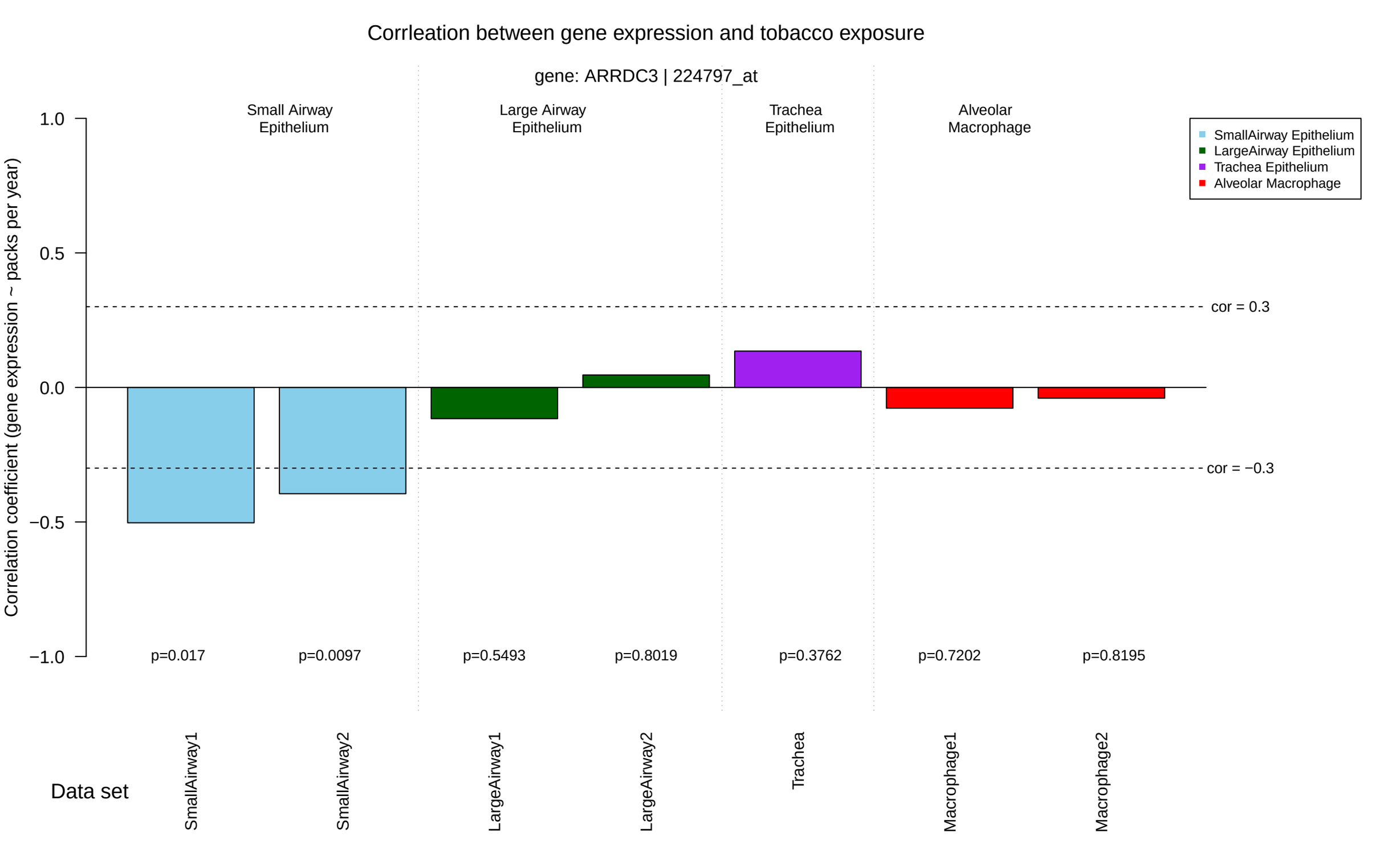

set: LargeAirway1 (p=0.1246)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5079)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3092)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5554)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.507)
ARRDC3 | 224797_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARRDC3 | 224797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.017
p=0.0097
p=0.5493
p=0.8019
p=0.3762
p=0.7202
p=0.8195
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2